

Edge Detection

- **Goal:** Identify points where intensity changes sharply, indicating boundaries of objects.
- Why Important?
 - Helps in object recognition, segmentation, feature extraction, tracking, etc.
- Edges represent significant structural information in an image.
- What's an Edge?
 - Edges are discontinuities in pixel intensity values.
- Goal of Edge Detection:
 - Approximate these transitions using gradient-based operators.

Common Edge Detection Methods

Method	Core Idea	Example Kernel
Sobel Operator	Measures gradient in x and y directions	$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$
Prewitt Operator	Similar to Sobel but simpler weights	$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$
Laplacian	Measures 2nd derivative (rate of change of gradient)	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
Canny Edge Detector	Multi-stage algorithm (noise removal + gradient + non-max suppression + hysteresis)	

Mathematical Background

Gradient Magnitude: $G = \sqrt{G_x^2 + G_y^2}$

Gradient Direction: $\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$

Given 3×3 pixel block:

$$I = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Compute G_x and G_y :

$$G_x = (-1)(10) + 0(10) + (1)(10) + (-2)(10) + 0(50) + (2)(10) + (-1)(10) + 0(10) + (1)(10) = 0$$

$$G_y = (-1)(10) + (-2)(10) + (-1)(10) + 0(10) + 0(50) + 0(10) + (1)(10) + (2)(10) + (1)(10) = -20$$

- Gradient Magnitude $G = 20$
- Gradient Direction is vertical as G_x is zero

Numerical Example (Using Sobel Operator)

Sobel Kernels:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Canny Edge Detector (Multi-Stage Process)

Goal: To achieve high accuracy, low false detection, and precise localization of edges.

Steps: `edges = cv.Canny(img,100,200)`

- Noise Reduction
 - Apply Gaussian blur to remove noise (e.g., 5×5 Gaussian filter).
- Gradient Calculation
 - Compute G_x , G_y using Sobel filters.
 - Find gradient magnitude and direction.
- Non-Maximum Suppression
 - Every pixel by keeping only local maxima in gradient direction.
- Hysteresis Thresholding
 - Classify edges as strong, weak, or non-edges using two thresholds.
- Produces a clean, continuous edge map.

